2514_DIT_ITSD001_Main EQP v2_Final(Apr 2025)

Question 1

(a)

Mr Lim is contemplating buying a small pre-owned car which is over 5 years old. After doing online research, he shortlisted the prices (in $'000) for ten such cars as shown below:

15	**21**	**21**	**28**	**32**	**43**	**45**	**50**	**56**	**68**

Compute the mean, median, mode, range and standard deviation. (Express your answer to 2 decimal places).
(10 marks)

Answer: Mean = 37.90, Median = 37.50, Mode = 21.00, Range = 53.00, Sample Standard Deviation = 17.27

考察知识点：第4章 Numerical Descriptive Measure。

解题步骤：

1. 先把数据按从小到大排列：
   15, 21, 21, 28, 32, 43, 45, 50, 56, 68
2. 计算平均数：

$$\bar{x} = \frac{15+21+21+28+32+43+45+50+56+68}{10} = \frac{379}{10} = 37.90$$

1. 计算中位数：
   共有 10 个数据，所以取第 5 个和第 6 个数据的平均值。

$$\text{Median} = \frac{32+43}{2} = 37.50$$

1. 计算众数：
   21 出现了 2 次，出现次数最多，所以众数是 21。
2. 计算极差：

$$\text{Range} = 68 - 15 = 53.00$$

1. 计算样本标准差：

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

先求平方和：

$$\sum(x_i - \bar{x})^2 = 2684.90$$

所以：

$$s = \sqrt{\frac{2684.90}{9}} = 17.27$$

(b)

James is planning to start a new business selling souvenirs at Orchard Road. He carried out a simple survey to find out the spending on souvenir by tourists. Data were gathered from a sample of 10 tourists at one souvenir shop at Far East Plaza.

Amount Spent ($)	**15**	**37**	**43**	**50**	**51**	**66**	**73**	**78**	**82**	**101**

Construct a stem and leaf diagram from the above data. Use stem unit = $10 and leaf unit = $1.
(6 marks)

Answer:

```
Stem | Leaf
 1   | 5
 3   | 7
 4   | 3
 5   | 0 1
 6   | 6
 7   | 3 8
 8   | 2
 9   |
10   | 1

Key: 5|1 = 51
```

考察知识点：第3章 Organising and Visualising Variables。

解题步骤：

1. 题目要求 stem unit = 10，leaf unit = 1，所以：
    - stem 表示十位数或更高位
    - leaf 表示个位数
2. 把每个数拆开：
    - 15 → 1 | 5
    - 37 → 3 | 7
    - 43 → 4 | 3
    - 50 → 5 | 0
    - 51 → 5 | 1
    - 66 → 6 | 6

- 73 → 7 | 3
- 78 → 7 | 8
- 82 → 8 | 2
- 101 → 10 | 1

3. 按 stem 从小到大排列，并把 leaf 排整齐，就得到上面的 stem-and-leaf diagram。

(c)

Mr Tan decided to examine the daily sales of new cars by Lee Motors over the last 20 days:

Number of cars sold	Number of days (Frequency)	Relative Frequency (%)	Class Mid-point
2 up to 5	5		
5 up to 8	6		
8 up to 11	4		
11 up to 14	3		
14 up to 17	2		
Total	20		

Based on the frequency distribution table above, calculate the relative frequencies (Express your answer to 2 decimal places) and class mid-points. Construct a histogram based on the frequency distribution table provided. Label all axes clearly.
(9 marks)

Answer:

Class Interval	Frequency	Relative Frequency (%)	Class Mid-point
2 up to 5	5	25.00	3.5
5 up to 8	6	30.00	6.5
8 up to 11	4	20.00	9.5
11 up to 14	3	15.00	12.5
14 up to 17	2	10.00	15.5
Total	20	100.00	

Histogram:

```
Frequency
6 |          ▇
5 | ▇        ▇
4 | ▇        ▇   ▇
3 | ▇        ▇   ▇   ▇
2 | ▇        ▇   ▇   ▇   ▇
1 | ▇        ▇   ▇   ▇   ▇
  +-----------------------------
    2-5      5-8   8-11 11-14 14-17
            Number of cars sold
```

考察知识点：第3章 Organising and Visualising Variables。

解题步骤：

1. relative frequency 的公式是：

$$\text{Relative Frequency} = \frac{\text{Frequency}}{20} \times 100\%$$

1. 逐组计算：
    - 5/20 × 100% = 25.00%
    - 6/20 × 100% = 30.00%
    - 4/20 × 100% = 20.00%
    - 3/20 × 100% = 15.00%
    - 2/20 × 100% = 10.00%
2. class mid-point 的公式是：

$$\text{Mid-point} = \frac{\text{lower limit} + \text{upper limit}}{2}$$

1. 逐组计算：
    - $(2+5)/2 = 3.5$
    - $(5+8)/2 = 6.5$
    - $(8+11)/2 = 9.5$
    - $(11+14)/2 = 12.5$
    - $(14+17)/2 = 15.5$
2. 画 histogram 时，柱子的高度就是频数。因为组距相同，所以每一组之间的柱子要连续画出来。

Question 2

(a)

Mr Lim has decided to recruit 10 new employees and there were 80 applicants for the jobs. He has tabulated the 80 applicants into the following contingency table based on their age group and degree level.

Age Group	Bachelor Degree Holders	Master Degree Holders	Total
20 to 29	11	6	17
30 to 39	19	8	27
40 to 49	17	10	27
50 and above	8	1	9
Total	55	25	80

Given that a job applicant is from the age group 30 to 39, what is the probability that the person is a master degree holder? What is the probability that a job applicant belongs to the age group 40 to 49 or the applicant is a bachelor degree holder? (Express answer to 3 decimal places)
(8 marks)

Answer: P(Master Degree Holder | 30 to 39) = 0.296
P(40 to 49 or Bachelor Degree Holder) = 0.813

考察知识点：第5章 Basic Probability。

解题步骤：

1. 第一问是条件概率：

$$P(\text{Master}|30 \text{ to } 39) = \frac{30 \text{ to } 39 \text{ and Master}}{30 \text{ to } 39 \text{ total}}$$

1. 由表可知：
    - 30 to 39 且 Master = 8
    - 30 to 39 总人数 = 27
2. 所以：

$$P(\text{Master}|30 \text{ to } 39) = \frac{8}{27} = 0.296$$

1. 第二问是“或”的概率，用加法公式：

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

1. 设：
    - A = 40 to 49
    - B = Bachelor Degree Holder

2. 代入表中的数据：

$$P(A) = \frac{27}{80}, \quad P(B) = \frac{55}{80}, \quad P(A \text{ and } B) = \frac{17}{80}$$

1. 所以：

$$P(A \text{ or } B) = \frac{27 + 55 - 17}{80} = \frac{65}{80} = 0.8125 \approx 0.813$$

(b)

Given a normal distribution with mean $\mu = 30$ and standard deviation $\sigma = 5$, what is $P(30 < X < 34)$? What is the probability a randomly selected sample of 25 customers has an average spending more than $32? (Express answer to 3 decimal place)
(8 marks)

Answer: P(30 < X < 34) = 0.288
P(\bar{X} > 32) = 0.023

考察知识点：第7章 Continuous Distributions and Normal Distributions；第8章 Sampling Distributions and Central Limit Theorem。

解题步骤：

1. 已知正态分布参数：

$$\mu = 30, \quad \sigma = 5$$

1. 先求 $P(30。
   先标准化：

$$Z = \frac{X - \mu}{\sigma}$$

1. 当 $X = 30$ 时：

$$Z = \frac{30 - 30}{5} = 0$$

1. 当 $X = 34$ 时：

$$Z = \frac{34 - 30}{5} = 0.8$$

1. 所以：

$$P(30 < X < 34) = P(0 < Z < 0.8)$$

1. 查标准正态分布表：

$$P(Z < 0.8) = 0.7881, \quad P(Z < 0) = 0.5000$$

1. 因此：

$$P(30 < X < 34) = 0.7881 - 0.5000 = 0.2881 \approx 0.288$$

1. 再求样本均值大于 32 的概率。样本量 $n = 25$。
2. 样本均值的标准误差：

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{25}} = 1$$

1. 计算 z 值：

$$Z = \frac{32 - 30}{1} = 2$$

1. 所以：

$$P(\bar{X} > 32) = P(Z > 2) = 1 - 0.9772 = 0.0228 \approx 0.023$$

(c)

In a jackpot game, player can spend $80 on an opportunity to push a button to control the displaying window. If the window shows cherry, they get $200. If the window shows lemon, they get $300. If the window shows coin, they get $500. If the window shows others, they get nothing. The probability of getting cherry is 1/8, the probability of getting lemon is 1/27. The probability of getting coin is 1/64. What amount can a player expect to win or lose by playing the game?
(9 marks)

Answer: The expected result is a loss of $36.08 per play.

考察知识点：第6章 Discrete Probability Distributions。

解题步骤：

1. 先计算每种情况的期望奖金额：

$$E(X) = 200\left(\frac{1}{8}\right) + 300\left(\frac{1}{27}\right) + 500\left(\frac{1}{64}\right) + 0$$

1. 分别计算：

$$200\left(\frac{1}{8}\right) = 25$$

1. 所以期望得到的奖金为：

$$25 + 11.1111 + 7.8125 = 43.9236$$

1. 但是玩一次要先花 80 元，所以净期望值为：

$$43.9236 - 80 = -36.0764$$

1. 负号表示亏损，因此玩家平均每次会亏损 36.08 元。

Question 3

(a)

A shop sells a particular layer cake. It is assumed that the daily sales of this type of cake is a Poisson variable with mean 5. Find the probability that on a particular day the shop sells:

(i) Exactly 6 cakes.
(ii) At least 2 cakes. (Correct your answers to 4 decimal places)
(10 marks)

Answer:
(i) P(X = 6) = 0.1462
(ii) P(X \ge 2) = 0.9596

考察知识点：第6章 Discrete Probability Distributions。

解题步骤：

1. 这是 Poisson distribution，已知平均数：

$$\lambda = 5$$

1. Poisson 公式：

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

1. 第(i)问，求恰好卖出 6 个蛋糕：

$$P(X = 6) = \frac{e^{-5}5^6}{6!}$$

1. 第(ii)问，求至少卖出 2 个蛋糕：
   先用补事件更方便：

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

1. 计算：

$$P(X = 0) = e^{-5}$$

1. 所以：

$$P(X \geq 2) = 1 - e^{-5} - 5e^{-5}$$

(b)

In a factory, on average, every one out of six earpieces is found to be defective. Twelve earpieces are selected at random. Find the probability that less than three of them will be defective. (Correct your answer to 4 decimal places)
(9 marks)

Answer: P(X < 3) = 0.6774

考察知识点：第6章 Discrete Probability Distributions。

解题步骤：

1. 这是二项分布问题。
   设：

$$n = 12, \quad p = \frac{1}{6}, \quad q = 1 - p = \frac{5}{6}$$

1. “less than three” 表示：

$$X = 0, 1, 2$$

1. 所以要算：

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

1. 用二项分布公式：

$$P(X = x) = \binom{12}{x} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{12-x}$$

1. 逐项计算并相加后得到：

$$P(X < 3) = 0.6774$$

(c)

State the level of measurement (Ratio, Ordinal or Nominal) for the variables: “Time spent revising for common test” and “Handphone number”.
(6 marks)

Answer: Time spent revising for common test = Ratio. Handphone number = Nominal.

考察知识点：第2章 Defining and Collecting Data。

解题步骤：

1. “Time spent revising for common test” 是时间变量。
   它有真实的 0，且可以比较倍数关系，所以属于 ratio。

2. “Handphone number” 只是号码标识，不代表大小、顺序或数量，所以属于 nominal。

Question 4

(a)

A survey of 20 teachers in a school found that the mean age of the teachers is 40.6 years old, with a sample standard deviation of 9.5 years. Find the 99% confident interval of the population mean. (Express your answer to 3 decimal places)
(9 marks)

Answer: The 99% confidence interval is (34.523, 46.677)

考察知识点：第9章 Confidence Interval and Point Estimation。

解题步骤：

1. 已知：

$$\bar{x} = 40.6, \quad s = 9.5, \quad n = 20$$

1. 因为总体标准差未知，所以使用 t 置信区间。
2. 自由度：

$$df = n - 1 = 19$$

1. 99% 置信水平对应：

$$t_{0.005,19} \approx 2.861$$

1. 标准误差：

$$SE = \frac{s}{\sqrt{n}} = \frac{9.5}{\sqrt{20}} = 2.125$$

1. 误差界：

$$ME = 2.861 \times 2.125 = 6.077$$

1. 置信区间：

$$40.6 \pm 6.077$$

(b)

An additional 20 teachers were surveyed. The mean age remained at 40.6 years and the sample standard deviation remained at 9.5 years. Find the 95% confident interval for the population mean age. (Express your answer to 3 decimal places.)
(9 marks)

Answer: The 95% confidence interval is (37.562, 43.638)

考察知识点：第9章 Confidence Interval and Point Estimation。

解题步骤：

1. 现在样本量变成 40，但均值和标准差不变：

$$\bar{x} = 40.6, \quad s = 9.5, \quad n = 40$$

1. 自由度：

$$df = 39$$

1. 95% 置信水平对应：

$$t_{0.025,39} \approx 2.023$$

1. 标准误差：

$$SE = \frac{9.5}{\sqrt{40}} = 1.503$$

1. 误差界：

$$ME = 2.023 \times 1.503 = 3.038$$

1. 置信区间：

$$40.6 \pm 3.038$$

(c)

The surveyor has made a mistake in calculating the standard deviation and the correct sample standard deviation has a smaller value of 7. How would this affect the width of the confidence interval?
(7 marks)

Answer: The confidence interval becomes narrower.

考察知识点：第9章 Confidence Interval and Point Estimation。

解题步骤：

1. 置信区间的宽度由误差界决定：

$$\text{Margin of Error} = t \cdot \frac{s}{\sqrt{n}}$$

1. 如果样本标准差变小，标准误差就会变小：

$$\frac{s}{\sqrt{n}} \downarrow$$

1. 标准误差变小，误差界也会变小。
2. 所以整个 confidence interval 会变窄。

END OF PAPER
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